



Library Insights: A Data Analysis Project using SQL

➤ Objective:

To analyze the library system to understand book lending trends, member behaviors, branch performance, and employee involvement in book issue/return processes.

➤ DataSet:

- **books.csv** – Details of books available in the library
- **branch.csv** – Information about library branches
- **employees.csv** – Staff members who manage books and memberships
- **issued_status.csv** – Information about which books were issued and by whom
- **members.csv** – Registered library members
- **return_status.csv** – Details about returned books

➤ Data Preparation & Transformation (Sample Tasks)

1. Create a column to calculate the total number of days a book was issued (return_date - issue_date).
2. Flag overdue books using a business rule (e.g., books kept for more than 14 days).
3. Create an "issue_month" or "return_month" column for time-based analysis.
4. Derive a "member_age_group" column using member date of birth or age (e.g., Teen, Adult, Senior).
5. Create a "book_status" column based on whether a book has been returned or not.
6. Classify members as "Frequent," "Moderate," or "Rare" borrowers based on their issue frequency.



➤ **Members**

1. Which members have borrowed the most books?
2. What is the average number of books borrowed per member?
3. How many active vs. inactive members are there?
4. What is the distribution of members across different age groups?
5. Which members frequently return books late?

➤ **Books**

6. What are the top 10 most issued books?
7. Which books are rarely borrowed?
8. What is the average issue duration of books?
9. Which book categories/genres are the most popular (if available)?
10. Are there books that are frequently overdue?

➤ **Branches**

11. Which branch has the highest number of issued books?
12. What is the average return delay per branch?
13. How do the branches compare in terms of active member engagement?
14. Which branch has the most overdue books?
15. Which branch has the largest collection of books?

➤ **Employees**

16. Which employees are handling the most number of book issues?
17. Which employees are responsible for delayed returns?
18. What is the employee-to-member interaction ratio?
19. Are there employees whose managed transactions have more overdue books?
20. Which employee has processed the highest number of returns?

➤ Time-based Trends

- 21. What is the trend of book issues per month?
- 22. In which month does the library see peak borrowing?
- 23. Are there seasonal spikes in book borrowing (e.g., summer holidays)?
- 24. What is the average issue duration per month?
- 25. How does return rate change over months or quarters?

➤ Return Behavior

- 26. What percentage of books are returned late?
- 27. What is the average delay in returning books?
- 28. How many books are currently not returned?
- 29. What's the average return duration by member type or branch?
- 30. Are certain books more likely to be returned late?